

Scaffolding Low-Resource STEM Education with Large Language Models

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Abstract

Low resource language learners can face difficulties understanding specialized English STEM terminology and concepts because translation alone may not provide sufficient contextual or educational support. The SLR and SSR identify fragmented approaches across multilingual translation, terminology handling, conceptual explanation, personalized learning, and LLM-based interaction. This research establishes a theoretical foundation using Task–Technology Fit as the primary Information Systems theory and Scaffolding Theory as a complementary educational theory. Guided by these foundations and the PIRQOA, three conceptual artefacts are proposed: core concepts defining LLM-based STEM scaffolding, a task-aligned model explaining relationships among language barriers, scaffolding, learner response, and STEM understanding, and a context-aware adaptive scaffolding framework describing how support should be structured and adjusted. Together, these artefacts provide theoretically grounded conceptual knowledge for aligning LLM capabilities with learner tasks and supporting terminology understanding, conceptual learning, and adaptive assistance for Burmese-speaking STEM learners.

Keywords: Educational Scaffolding, Large Language Models, Low Resource Languages, STEM Terminology, Multilingual Learning, Burmese Language

1. Theoretical Foundation Based on Existing Artefacts

The theoretical foundation of this research combines a primary Information Systems theory, a complementary educational theory, and supporting concepts and processes identified through the SLR and SSR. The literature shows that low resource language barriers, specialized terminology, conceptual explanation, personalized learning, and LLM-based interaction are often addressed separately. The PIRQOA therefore requires terminology and language support, conceptual explanation, and structured and adaptive scaffolding. These requirements provide the basis for selecting Task–Technology Fit (TTF) as the primary Information Systems theory and Scaffolding Theory as a complementary educational theory.

Task–Technology Fit (TTF) provides the primary theoretical foundation. Goodhue and Thompson (1995) propose that information technology is more likely to support individual performance when its capabilities match the requirements of the task. In this research, the learner's task extends beyond translating English into Burmese. Learners need to identify and understand specialized English STEM terminology, receive

context-sensitive language support, understand underlying concepts, and access appropriate learning assistance. The technology is an LLM-based educational Information System capable of providing multilingual and interactive support. TTF is therefore relevant because it provides a basis for aligning technological capabilities with the requirements of STEM-learning tasks.

The SLR provides empirical justification for this application of TTF. Native-language interaction can reduce difficulties associated with English terminology and support conceptual understanding (Athukorala & De Silva, 2025). Specialized terminology also needs to be distinguished from ordinary vocabulary before appropriate support can be provided (Tran et al., 2026), while direct translation may produce unfamiliar or misleading technical terminology (Kleidermacher & Zou, 2026). Structured interaction, feedback, and adaptive assistance are also relevant educational capabilities (Kuzu, 2026; Nair et al., 2026). TTF therefore establishes task–capability alignment as a central design principle: terminology support should address terminology-related difficulties, conceptual explanations should support understanding beyond translation, and adaptive assistance should respond to learner needs.

Scaffolding Theory provides complementary knowledge about how educational assistance should behave. Although it originates from educational research rather than Information Systems, it is appropriate because educational scaffolding is a central function of the intended system. Wood et al. (1976) describe scaffolding as assistance that enables learners to perform tasks beyond their current independent capability. TTF explains whether technological capabilities fit the learner's task, whereas Scaffolding Theory explains how assistance should be structured and adjusted.

The SLR further supports this theory selection. Kuzu (2026) demonstrates structured multilingual LLM-supported learning involving prompt formulation, response evaluation, and explanation refinement. Nair et al. (2026) similarly demonstrates bilingual interaction, personalized feedback, and structured learning activities. These findings support guided and learner-responsive interaction rather than unrestricted access to a general-purpose LLM. Scaffolding Theory therefore informs the use of explanations, examples, prompts, feedback, and adaptive guidance.

Several additional concepts and processes identified through the SLR provide operational knowledge without requiring further kernel theories. Automatic Term Extraction informs the identification of specialized STEM terminology (Tran et al., 2026). Context-sensitive translation informs decisions about whether terminology should be translated, retained in English, presented bilingually, or accompanied by explanation (He et al., 2025; Kleidermacher & Zou, 2026). Multilingual prompt engineering provides mechanisms for guiding multilingual LLM behavior through structured, bilingual, few-shot, or cross-lingual prompting (Vatsal et al., 2026).

Table 1

Theoretical and Conceptual Foundations for the Proposed Artefacts

Foundation	Type	Contribution to this research
Task–Technology Fit	Primary IS theory	Aligns LLM capabilities with learner tasks and support requirements.
Scaffolding Theory	Complementary educational theory	Defines how explanations, guidance, feedback, and adaptive assistance should support learners.
Automatic Term Extraction	Existing concept/process	Supports identification of specialized STEM terminology.
Context-sensitive translation	Existing concept/process	Informs translation, terminology preservation, bilingual presentation, and explanation.
Multilingual prompt engineering	Existing process	Guides multilingual translation, explanation, and interaction behavior.

Together, these foundations support the proposed conceptual artefacts. TTF establishes alignment between learner tasks and technological capabilities, while Scaffolding Theory defines the educational behavior of that support. The supporting concepts provide operational knowledge for terminology identification, multilingual assistance, and LLM interaction.

Following Gregor and Hevner's (2013) Knowledge Contribution Framework, this research is positioned primarily as an Improvement contribution. The problem is established, and existing studies already provide capabilities for translation, terminology identification, explanation, personalized learning, and LLM interaction. However, these capabilities remain fragmented. The proposed artefacts therefore extend existing design knowledge by integrating task–technology alignment, educational scaffolding, terminology-sensitive language support, conceptual explanation, and adaptive interaction into a coherent approach for low resource STEM learning.

2. Newly Proposed Conceptual Artefacts

Drawing on the theoretical foundation, SLR, SSR, and PIRQOA, three complementary conceptual artefacts are proposed. Artefact 1 defines the concepts that constitute LLM-based STEM scaffolding, Artefact 2 models the relationships among its main constructs, and Artefact 3 specifies the structured and adaptive process through which support should operate. Together, they progress from definition to relationship to behavior and provide conceptual knowledge that can guide subsequent system design.

2.1. Conceptual Artefact 1: LLM-Based STEM Scaffolding Concepts

The first artefact defines LLM-Based STEM Scaffolding and its core concepts. The SLR shows that terminology identification, multilingual support, conceptual explanation, personalized assistance, and structured LLM interaction are addressed across different research streams but remain insufficiently integrated.

In this research, LLM-Based STEM Scaffolding is defined as the structured use of LLM capabilities to identify specialized STEM terminology, provide context-sensitive language support, explain underlying concepts, and adjust educational assistance according to learner interaction and task requirements. It consists of five core concepts.

STEM Terminology Identification refers to recognizing specialized domain terms requiring language or conceptual support. It is informed by Automatic Term Extraction, which distinguishes specialized terminology from ordinary vocabulary (Tran et al., 2026).

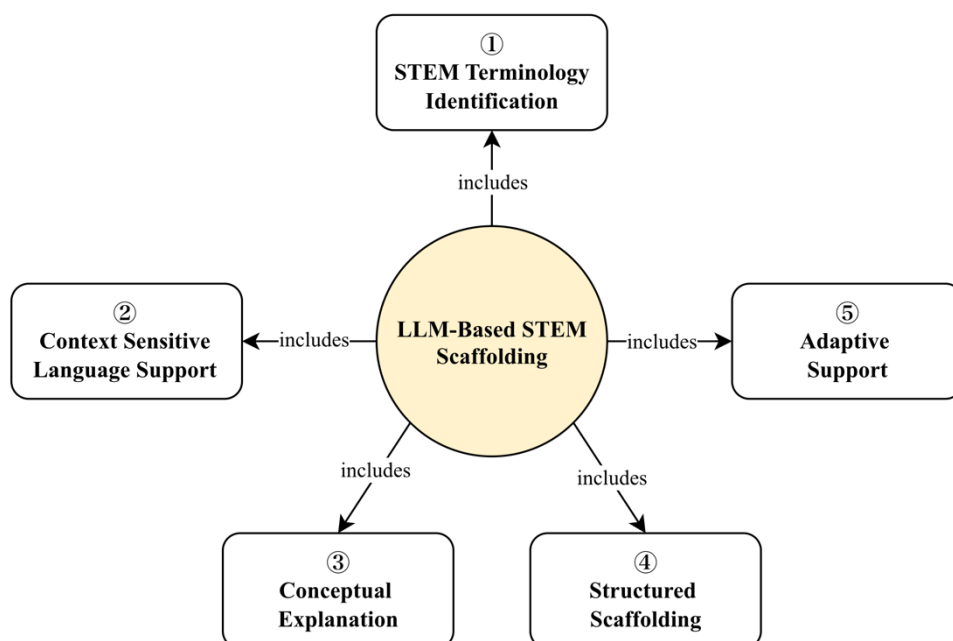
Context-Sensitive Language Support refers to selecting appropriate assistance according to the technical and linguistic context. Support may involve Burmese translation, preservation of the English term, bilingual presentation, or additional explanation. This reflects evidence that direct translation can reduce familiarity or distort technical meaning (He et al., 2025; Kleidermacher & Zou, 2026).

Conceptual Explanation extends beyond lexical translation by explaining what a STEM concept means. Native-language and structured explanatory support can improve understanding of technical concepts (Athukorala & De Silva, 2025; Kuzu, 2026).

Structured Scaffolding refers to organized assistance such as explanations, examples, prompts, and guidance rather than unrestricted chatbot interaction. It is grounded in Scaffolding Theory (Wood et al., 1976) and supported by structured multilingual LLM interaction identified by Kuzu (2026).

Figure 1

Conceptual Structure of LLM-Based STEM Scaffolding



Adaptive Support refers to adjusting the level or type of assistance according to learner interaction and needs. Personalized feedback and structured learning support demonstrated by Nair et al. (2026) provide evidence for this learner-responsive approach.

Figure 1 positions LLM-Based STEM Scaffolding as the overarching concept and the five concepts as its defining elements. The relationship “includes” is used because the concepts constitute its meaning rather than represent causal relationships. Artefact 1 therefore addresses PIRQOA requirements concerning terminology support, conceptual explanation, and structured and adaptive assistance.

2.2. Conceptual Artefact 2: Task-Aligned LLM Scaffolding Model

The second artefact models how four major constructs are related: Low Resource Language Barrier, LLM-Based Scaffolding, STEM Understanding, and Learner Response.

First, the Low Resource Language Barrier hinders STEM Understanding. Difficulties with English technical terminology can limit access to technical knowledge, while native-language interaction can support conceptual understanding (Athukorala & De Silva, 2025).

Second, LLM-Based Scaffolding reduces the Low Resource Language Barrier. Terminology identification and context-sensitive language support are intended to make specialized English STEM content more accessible. The term reduces is used rather than eliminates because language barriers may remain.

Third, LLM-Based Scaffolding supports STEM Understanding. Translation alone may not provide sufficient conceptual understanding; specialized terminology can require contextual explanation or preservation (Athukorala & De Silva, 2025; Kleidermacher & Zou, 2026). Scaffolding therefore extends beyond translation through explanation and structured learning support.

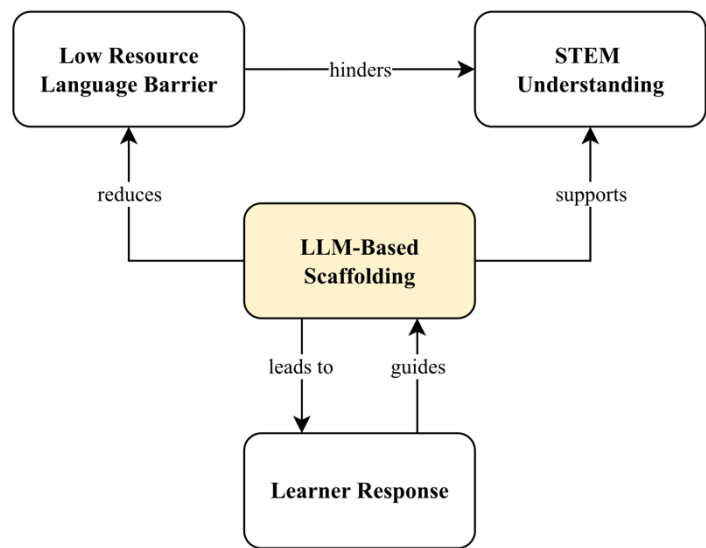
Fourth, LLM-Based Scaffolding leads to Learner Response through questions, answers, clarification requests, or other interaction.

Finally, Learner Response guides LLM-Based Scaffolding, creating a feedback loop in which subsequent assistance can be adjusted. This relationship is consistent with Scaffolding Theory (Wood et al., 1976) and structured LLM-supported learning in which learners evaluate and refine responses (Kuzu, 2026).

TTF provides the primary theoretical grounding for this model because effective support depends on alignment between learner-task requirements and technological capabilities (Goodhue & Thompson, 1995). Scaffolding Theory complements TTF by explaining why assistance should respond to learner interaction.

Artefact 2 therefore represents both the supportive role of scaffolding and its adaptive feedback relationship.

Figure 2
Task-Aligned LLM Scaffolding Model



2.3. Conceptual Artefact 3: Context-Aware Adaptive STEM Scaffolding Framework

The third artefact transforms the concepts and relationships from Artefacts 1 and 2 into a structured, iterative process.

The process begins with Identify STEM Terminology, where specialized terms are distinguished from ordinary vocabulary using knowledge from Automatic Term Extraction (Tran et al., 2026).

The identified terminology then determines Interpret Technical Context, which considers the meaning of the term within its STEM discipline and learning context rather than treating it as an isolated word (He et al., 2025).

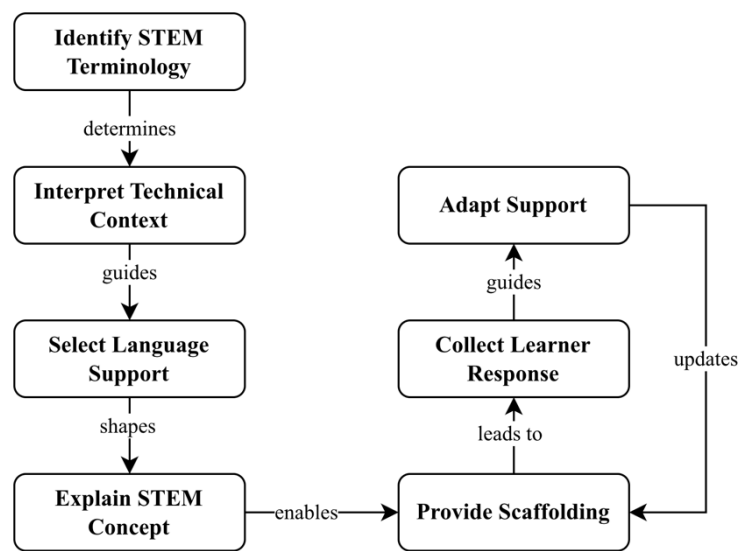
Technical context guides Select Language Support. Depending on the term and context, support may involve Burmese translation, English-term preservation, bilingual presentation, or contextual explanation (He et al., 2025; Kleidermacher & Zou, 2026).

The selected support shapes Explain STEM Concept, which moves beyond translation to clarify conceptual meaning and, where appropriate, provide examples or clarification (Athukorala & De Silva, 2025; Kuzu, 2026).

Conceptual explanation enables Provide Scaffolding, where structured assistance may include explanations, examples, questions, hints, or guidance. This stage is grounded in Scaffolding Theory (Wood et al., 1976) and supported by structured LLM interaction identified in the SLR (Kuzu, 2026).

Providing scaffolding leads to Collect Learner Response. The learner's response may indicate understanding, uncertainty, or the need for further support. This response then guides Adapt Support, where subsequent assistance may be increased, reduced, clarified, or changed (Nair et al., 2026). Adapted support updates the scaffolding provided, creating an iterative feedback process.

Figure 3
Context-Aware Adaptive STEM Scaffolding Framework



The framework is therefore both context-aware and adaptive. TTF requires each stage to correspond to an identifiable learner task or support requirement (Goodhue & Thompson, 1995), while Scaffolding Theory informs the progression and adjustment of educational assistance (Wood et al., 1976). Automatic Term Extraction, context-sensitive translation, and multilingual prompt engineering provide additional implementation-oriented knowledge (He et al., 2025; Kleidermacher & Zou, 2026; Tran et al., 2026; Vatsal et al., 2026) without turning the conceptual framework into a software architecture.

2.4. Alignment of the Conceptual Artefacts with PIRQOA

The three artefacts respond directly to the problems, issues, requirements, research questions, and objectives established through the PIRQOA. Table 2 summarizes this traceability.

Table 2
Alignment of Proposed Conceptual Artefacts with PIRQOA

PIRQOA Element	Artefact 1: LLM-Based STEM Scaffolding Concepts	Artefact 2: Task-Aligned LLM Scaffolding Model	Artefact 3: Context-Aware Adaptive STEM Scaffolding Framework
Problems	Defines the terminology, language-support, conceptual-explanation, structured-scaffolding, and adaptive-support concepts needed to address the identified learning difficulties.	Models how the Low Resource Language Barrier hinders STEM Understanding and how LLM-Based Scaffolding can reduce the barrier and support understanding.	Integrates terminology, language, conceptual, and adaptive support into a structured process for addressing the identified problems.
Issues	Defines the core forms of support needed for terminology and language support, conceptual learning support, and structured/adaptive scaffolding.	Explains the relationships between language barriers, scaffolding, STEM understanding, and learner response, clarifying why these issues are connected.	Specifies how terminology support, conceptual explanation, learner response, and adaptation should operate together.
Requirements	Defines the capabilities required for terminology identification, context-sensitive Burmese support, conceptual explanation, and adaptive assistance.	Provides the conceptual rationale for why scaffolding should reduce language barriers, support STEM understanding, and respond to learner interaction.	Specifies the process for identifying terminology, interpreting context, selecting language support, explaining concepts, providing scaffolding, and adapting support.
Research Questions	Supports RQ1 by defining terminology and language-support concepts, RQ2 through Conceptual Explanation, and RQ3 through Structured Scaffolding and Adaptive Support.	Supports RQ1 indirectly by modelling the language barrier that terminology and language support are intended to reduce, RQ2 by modelling how scaffolding supports STEM understanding, and RQ3 through the learner-response feedback relationship.	Supports RQ1 through the terminology-support process, RQ2 through the explanation and scaffolding stages, and RQ3 through the learner-response and adaptation cycle.
Objectives	Provides the conceptual definitions needed to develop terminology support, concept explanation, and adaptive scaffolding approaches.	Provides the relationship model needed to examine how scaffolding relates to language barriers, STEM understanding, and learner response.	Provides the process framework that can guide development and later evaluation of terminology support, conceptual explanation, and adaptive scaffolding.

Collectively, the three artefacts provide complementary conceptual contributions. Artefact 1 defines what LLM-based STEM scaffolding consists of; Artefact 2 explains how its main constructs relate, and Artefact 3 specifies how the support should operate. This progression from definition to relationship to behavior provides a theoretically grounded foundation for subsequent system artefacts while remaining independent of any specific LLM, interface, API, or implementation architecture.

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